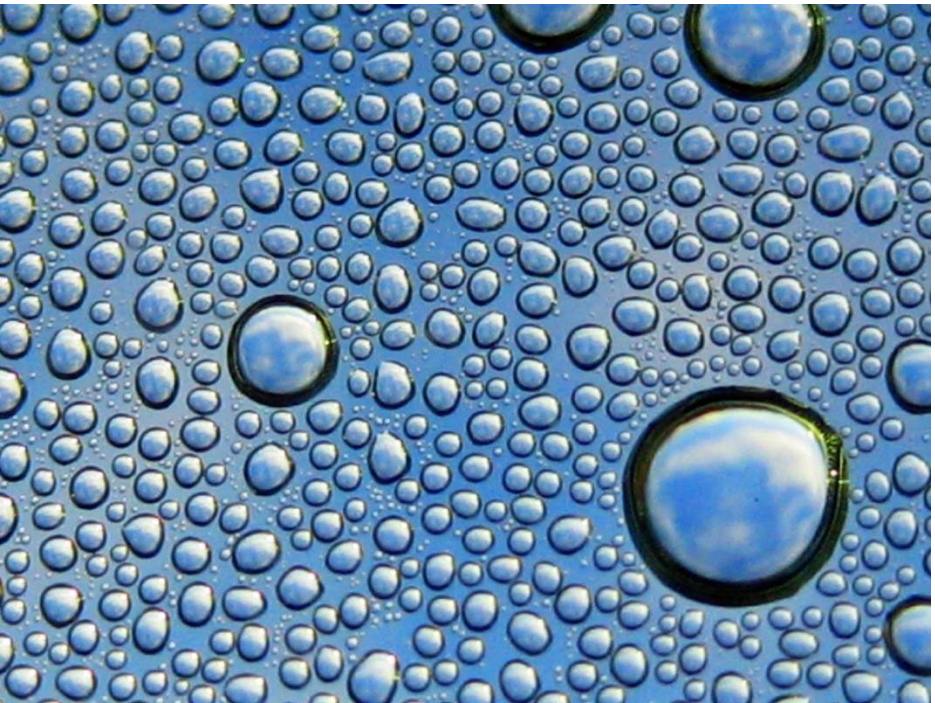
Structure & Properties of Water, Acid ,Base, Buffer,Handerson Haselbach Equation





STRUCTURE OF WATER

UNIVERSAL SOLVENT

DENSITY

COHESION

POLAR MOLECULE

SPECIFIC HEAT

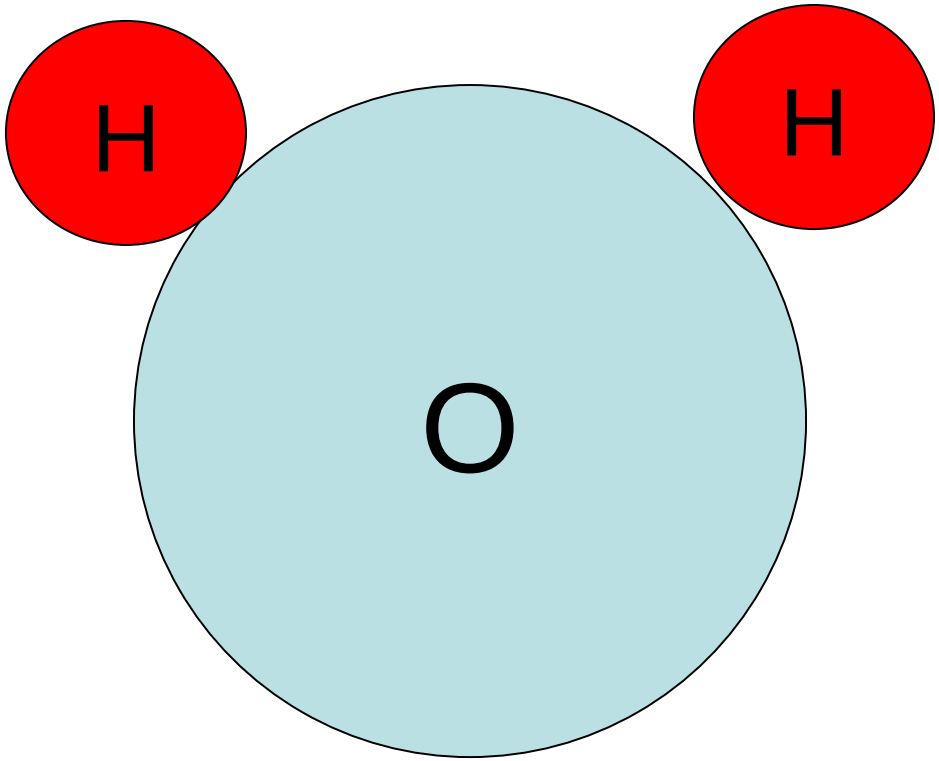
ADHESION

CAPILLARY ACTION

SURFACE TENSION

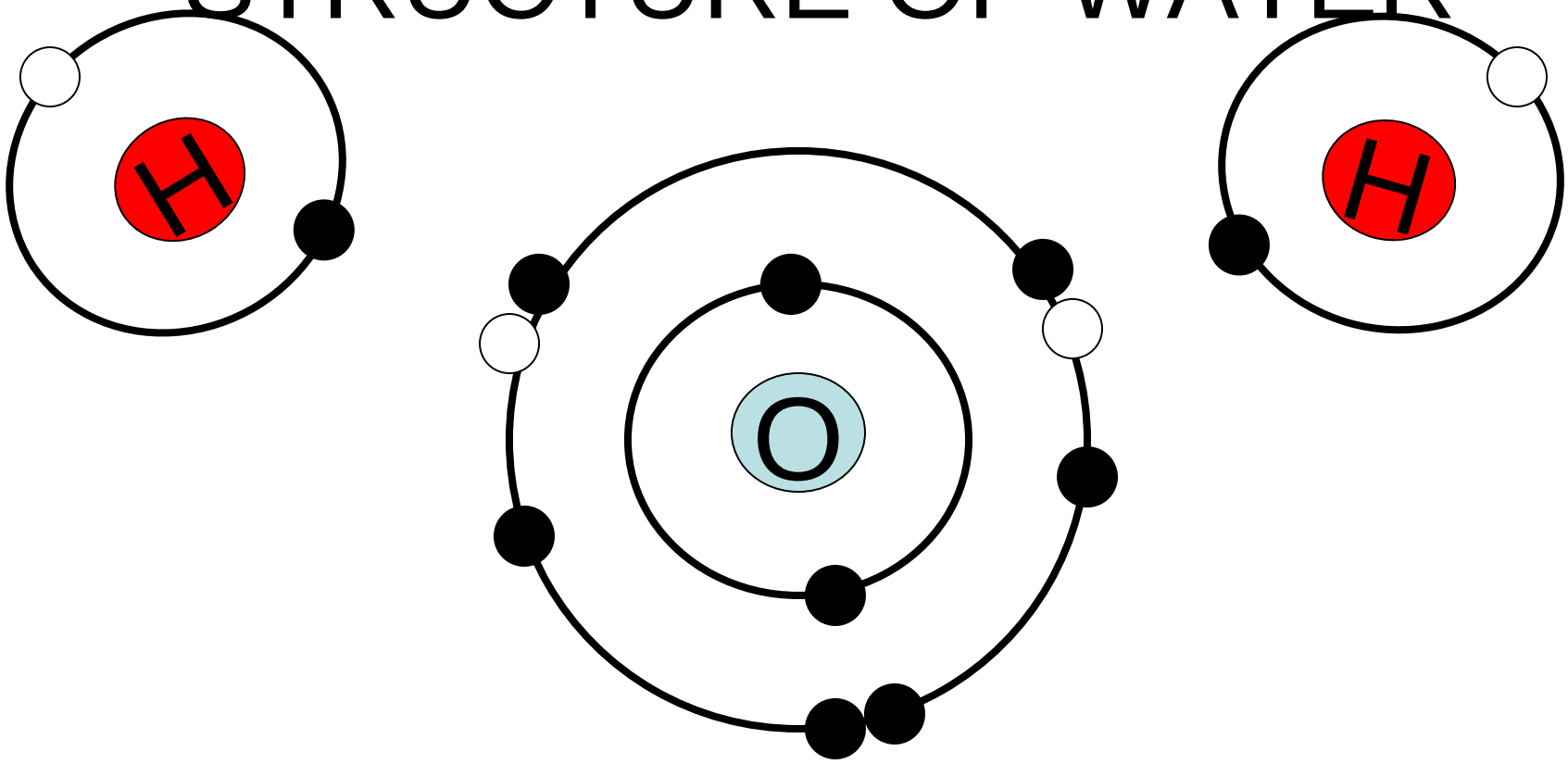
BUOYANCY

STRUCTURE OF WATER



1 molecule of water is made up of 2 hydrogen atoms bonded with 1 oxygen atom

STRUCTURE OF WATER



The bond that forms water is a <u>covalent bond</u>

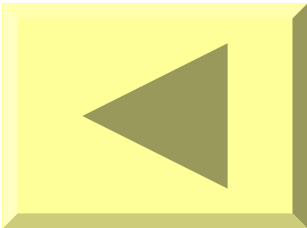
CHEMICAL FORMULA

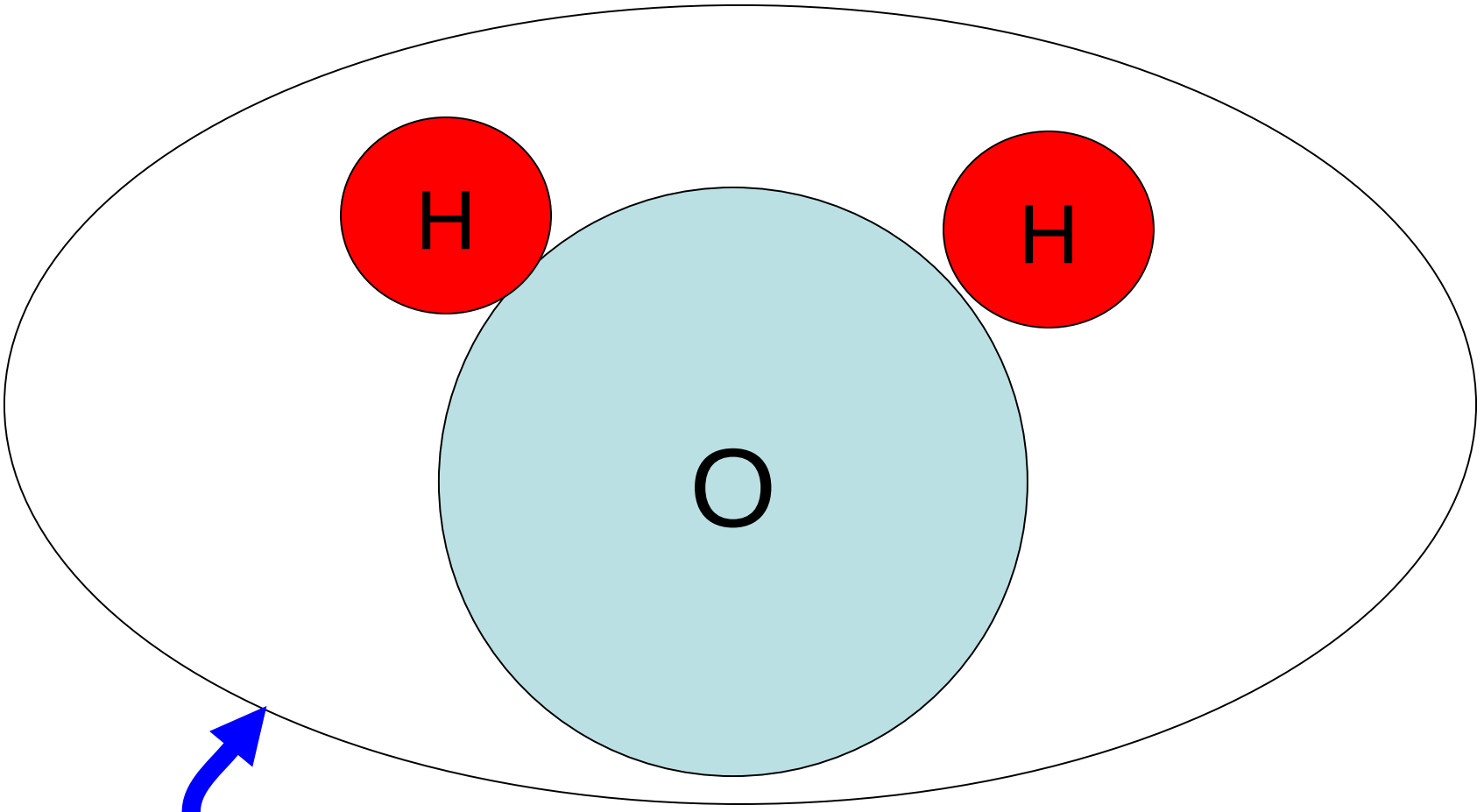
H_2O = RIGHT

H^2O = WRONG

H2O = WRONG

2HO = WRONG





1 molecule of water

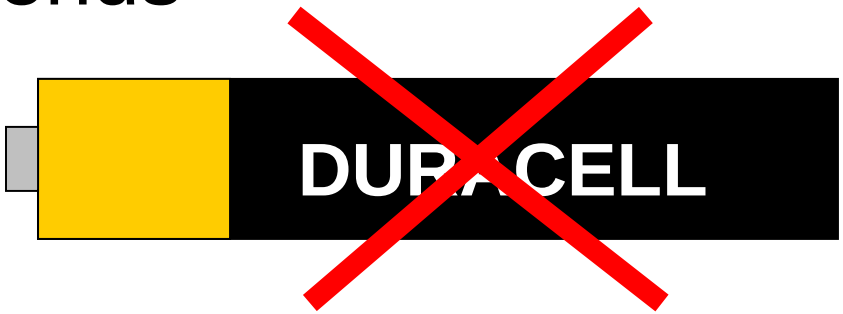
POLAR MOLECULE

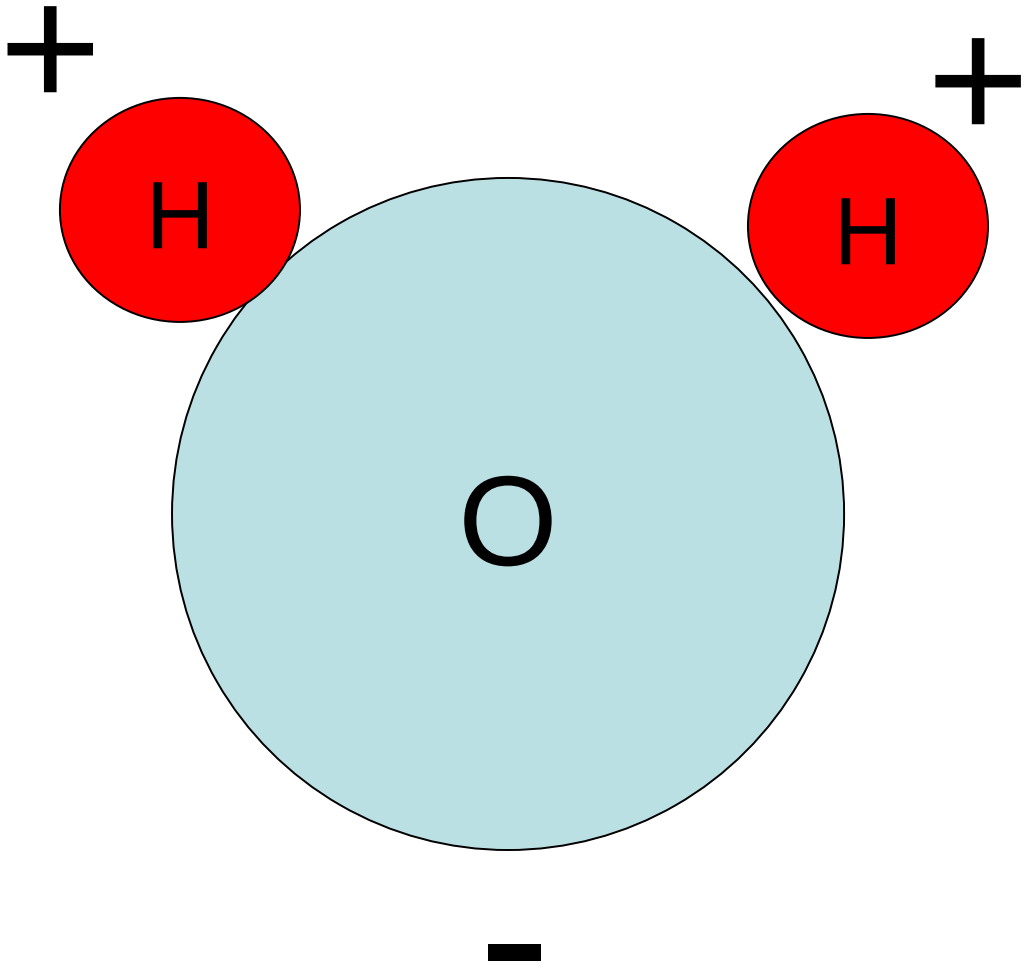
- A molecule that has electrically charged areas.
- Also called a polar substance or polar compound.



NON-POLAR SUBSTANCES

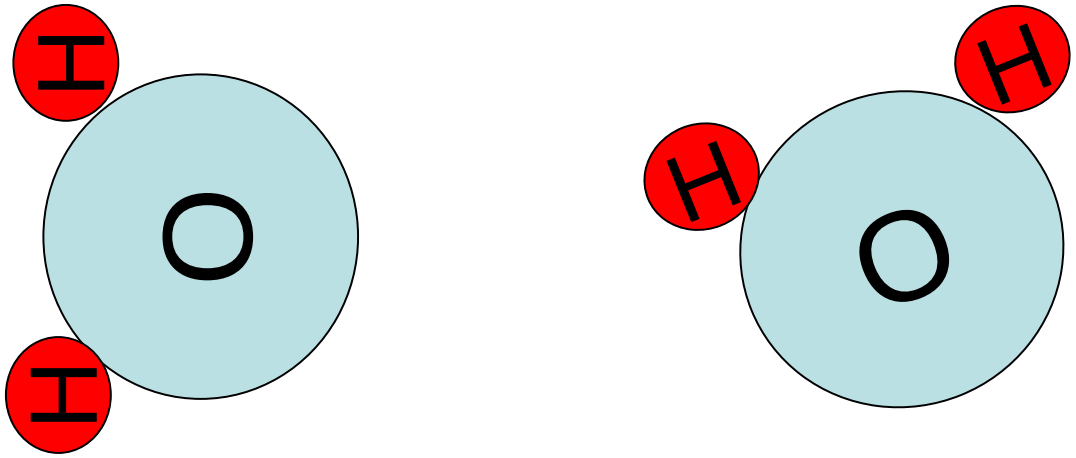
- Non-polar substances do not have charged ends




+
+
H
H
O
-

POLAR MOLECULE

The positive hydrogen ends of 1 molecule are attracted to the negative end of the oxygen of another molecule.



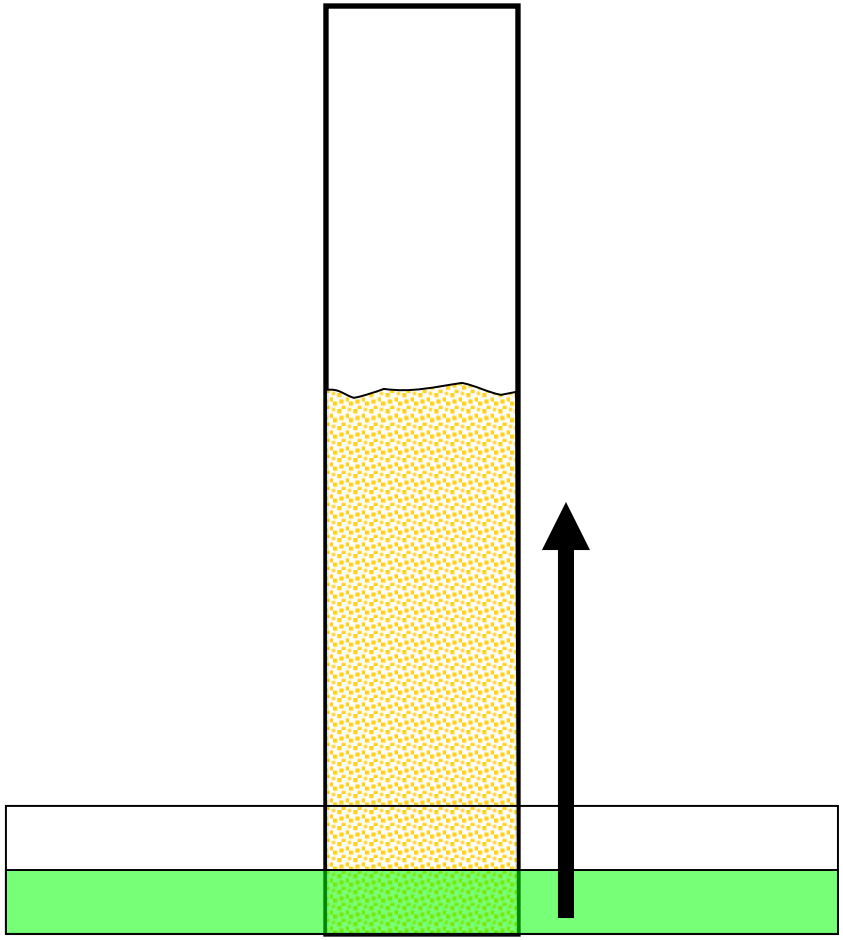
POLAR MOLECULE

THE PROPERTIES OF:

Cohesion, adhesion, specific heat, universal solvent, capillary action, surface tension

ALL HAPPEN BECAUSE WATER IS A POLAR SUBSTANCE!

CAPILLARY ACTION



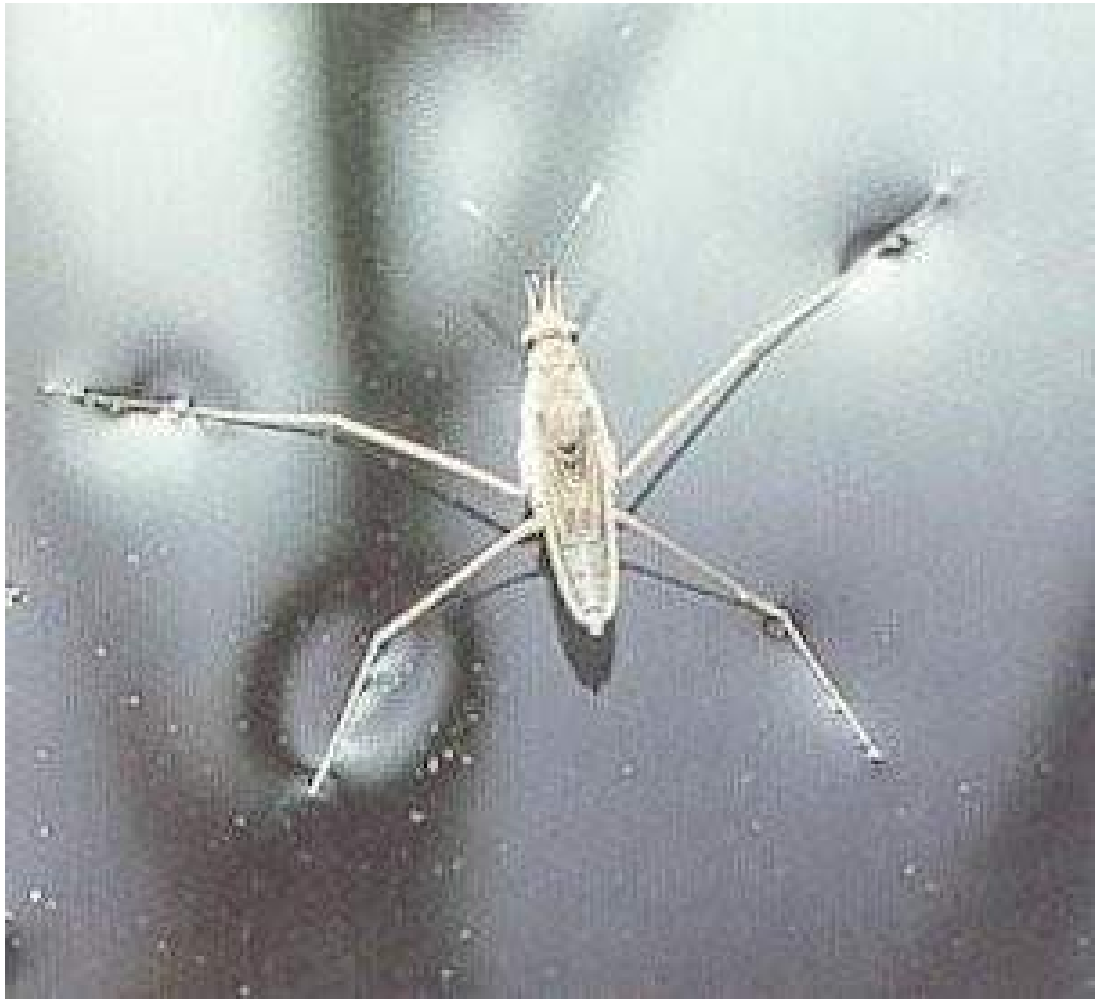
CAPILLARY ACTION

- The combined force of attraction among water molecules and with the molecules of the surrounding material.
- Cohesion + Adhesion

SURFACE TENSION



SURFACE TENSION



SURFACE TENSION



SURFACE TENSION

- The tightness across the surface of water that is caused by the polar molecules pulling on one another.
- Makes the surface act like a solid

UNIVERSAL SOLVENT



UNIVERSAL SOLVENT

- This is a nickname given to water.
- Water dissolves LOTS of things.
- Remember:

  SOLVENT = the thing doing the dissolving

  SOLUTE = the thing that dissolves away

- Other polar molecules can be dissolved by water.

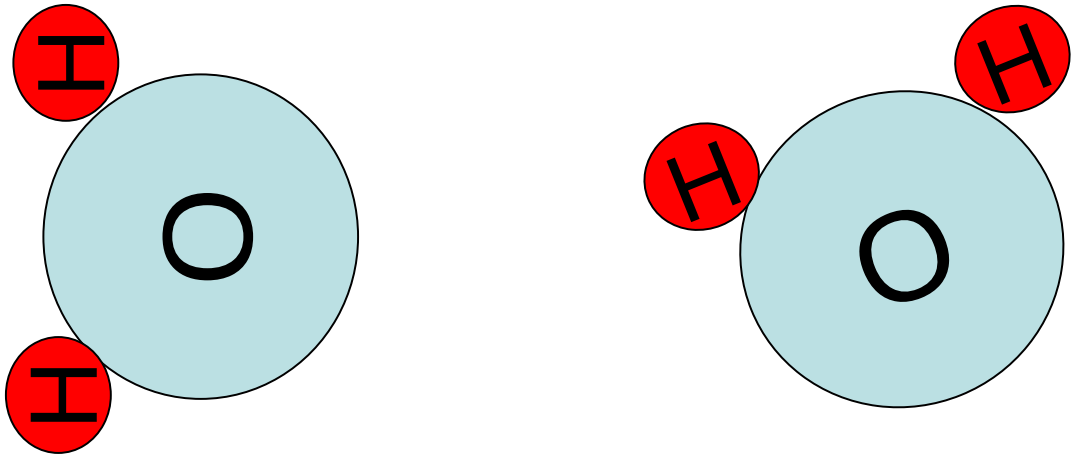
Universal Solvent

Why is water so good at dissolving things?

Because water is a polar molecule and is shaped like a wedge, it is able to break up substances into smaller pieces (dissolve).

COHESION

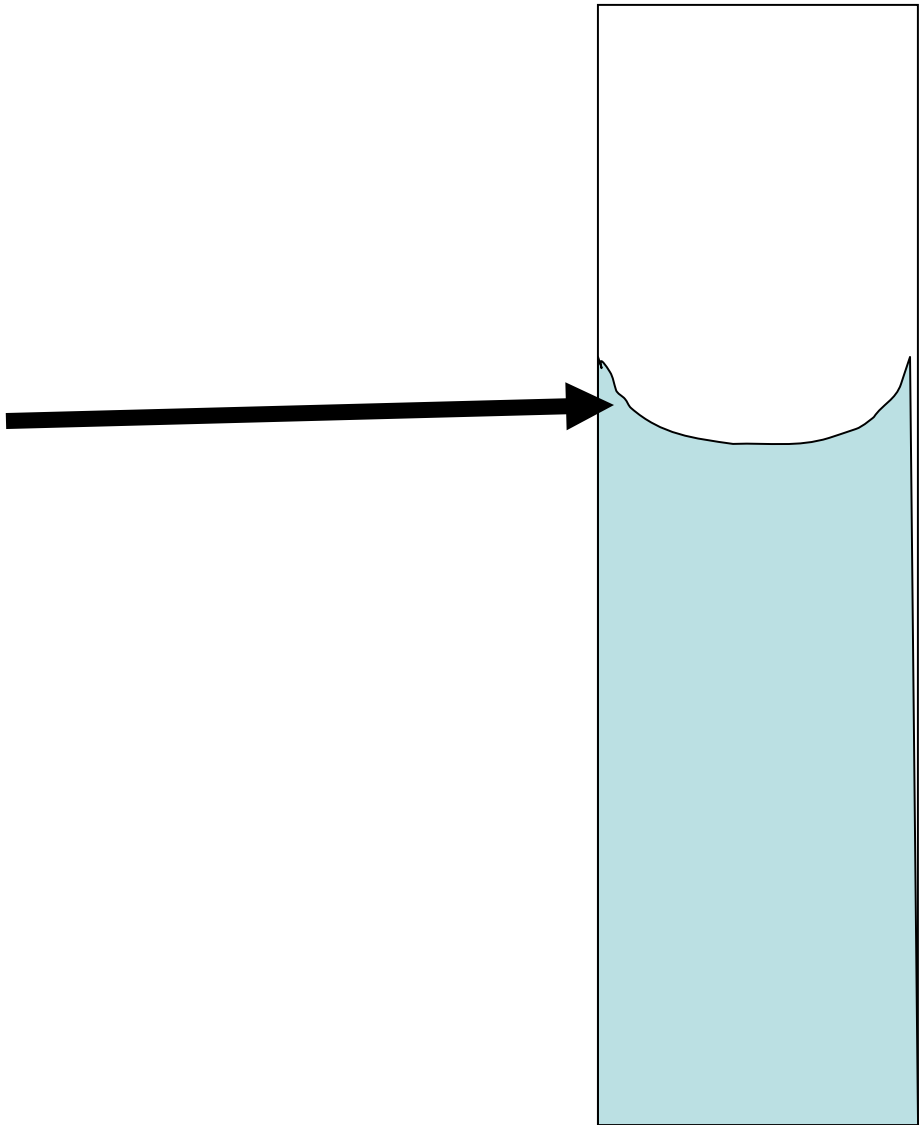
- Because water is a polar molecule, it is attracted to itself.
- **Cohesion** – is the attractive force between water molecules.



ADHESION

- Because water is a polar molecule, it is attracted to other substances.
- **Adhesion** – occurs when molecules of water are attracted to other substances.

ADHESION

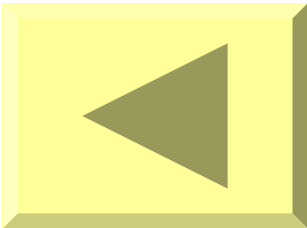


COHESION

Water molecules like to stick to each other.

ADHESION

Water molecules like to stick to other things.

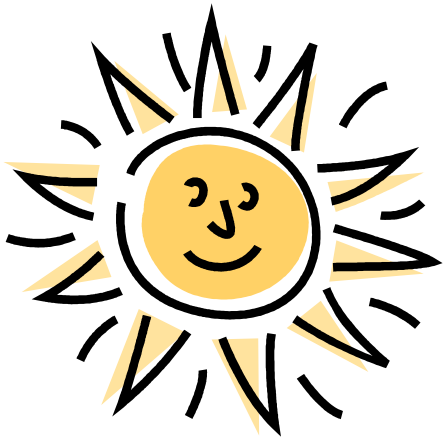


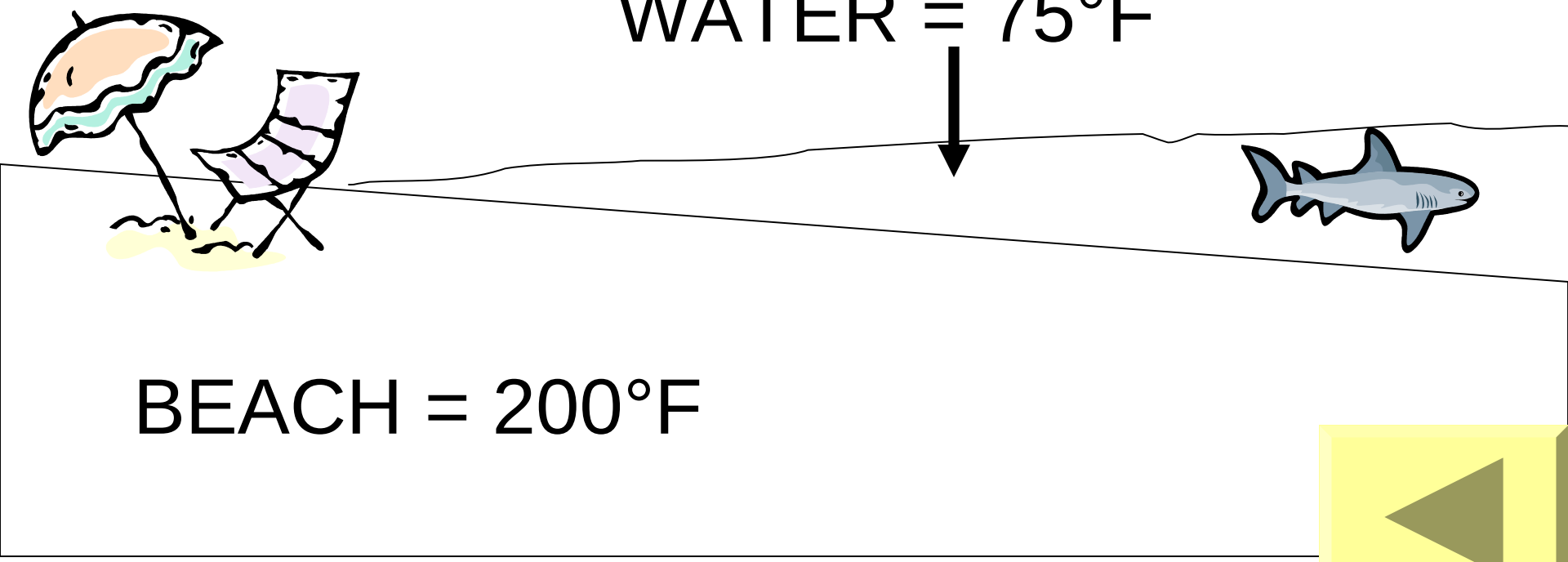
SPECIFIC HEAT

- Specific Heat = the amount of energy needed to increase the temperature of something 1 degree C.
- Water has a really HIGH specific heat
- That means it takes a lot of energy for water to increase its temperature.
- This is because of the STRONG ATTRACTION between water molecules.

  It's POLAR!

SPECIFIC HEAT





Acid dissociation constant

– The general ionization of an acid is as follows:

$$HA \longleftrightarrow H^+ + A^-$$

So the acid dissociation constant is as follows:

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

There are many orders of magnitude spanned by K_a values, so pK_a is used instead:

$$pK_a = -\log_{10} K_a$$

The larger the value of the pK_a, the smaller the extent of dissociation.

pK_a <2 is a strong acid

Henderson-Hasselbalch Equation

- **Describes the dissociation of a weak acid in the presence of its conjugate base**
  - The general ionization of a weak acid is as follows:

$$HA \longleftrightarrow H^+ + A^-$$

So the acid dissociation constant is as follows:

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

Rearranging this expression in terms of the parameter of interest [H+] gives the following:

$$[H^+] = \frac{K_a [HA]}{[A^-]}$$

Henderson-Hasselbalch Equation

Take the log of both sides:

$$\log[H^+] = \log K_a + \log \frac{[HA]}{[A^-]}$$

Change the signs and define pK_a as $-\log K_a$:

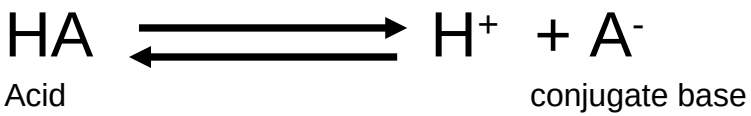
$$pH = pK_a - \log \frac{[HA]}{[A^-]}$$

or

$$\mathbf{pH = pK_a + \log \frac{[A^-]}{[HA]}}$$

Buffers

- **Buffers are solutions that resist changes in their pH as acid (H+) or base (OH-) is added**.
- Typically, buffers are composed of a weak acid and its conjugate base.
- Acids = Proton (H+) donors
- Bases = Proton Acceptors



- Acids and their conjugate bases are in equilibrium. Equilibria are related to the properties of the reactants and products, so for weak acids, the tendency to give up its proton determines its buffering property
- The tendency to ionize can be put in an equilibrium equation

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

- A solution of a weak acid that has a pH near to its pK_a has an equivalent amounts of conjugate base and weak acid.
- **Typically a weak acid is in its useful buffer range within 1 pH unit of its pK_a.**

Titration curve for the bicarbonate buffer system

